

Stratus Banks on Fault-Tolerant Systems' Success

Firm Says Units Lower Downtime

By Peter Bochner
CBN Staff

NATICK, Mass. — Expecting all markets to become solution-oriented, Stratus Computer Inc. eventually plans to corral hundreds of software houses to write applications software for its fault-tolerant computers.

Stratus software houses will collect a fee each time their software plays a substantial role in a system sale, according to Peter Kastner, Stratus' manager of marketing development. Stratus will give software houses a discount on a system, but will not require them to buy a computer, he added.

While Stratus' main thrust will be Fortune 1000 companies, "any company starting out that ignores OEMs is ignoring a big opportunity," Kastner told CBN last week. OEMs in particular will warm to fault-tolerant systems, he said, because they are usually the first point of contact for a customer whose computer has gone down. Accordingly, Stratus has end-user, software house and two-year OEM contracts.

Stratus recently signed an agreement for Olivetti to distribute its products in England, France and Italy. But Kastner pointed out the contract is also an OEM agreement for Olivetti. Overall, the contract could be worth up to \$40 million by 1985, he noted.

Stratus has released two software tools (Fortran 77 and Pascal) and five applications: Tranpro transaction processing system with network support; Forms Management System to create VDT forms; 3270 Terminal Support to connect IBM-

compatible VDTs to a Stratus/32; 3270 Emulator to let Stratus application programs interactively access an IBM-compatible host; and X.25 Networking for communications between application programs on systems from Stratus or other vendors.

Although many people consider Tandem Computers Inc. to be Stratus' main competitor, Kastner said he feels Stratus' transaction processing thrust

panies such as DG have not yet come out with fault-tolerant computers because of the difficulty of making them compatible with existing products. Hence, fault-tolerant offerings have been the domain of start-ups, but de Castro warned that start-ups often have problems getting out second-generation products.

Stratus' second generation is at least a year away. "We will do our best to maintain com-

of code to handle different architectures," Kastner noted. Should Stratus designers want a faster, more powerful CPU, they could protect the users' software investment by merely changing the code generator. Users could then recompile their programs on the new Stratus product.

Each Stratus board is capable of checking itself for errors; the concept of pairing boards off so as to present two CPUs to the

Powerful, low-cost computers such as Stratus', Kastner said, may reverse the trend of decentralized processing in the office somewhat.

Not that microcomputer workstations are going to disappear, "but you reach the point where it's a lot more economical to have the peripherals and memory for each micro not in the micro itself but in a central computer.

"You'd have the same response time, but the efficiencies of shared storage and higher storage."

Such a centralized minicomputer would also be capable of consolidating, for example, VisiCalc entries from many persons, something offline mainframes could not accomplish, he added. Kastner, a former analyst at Arthur D. Little, said he thinks the fault-tolerant market is blossoming from a niche market to a cross market. "As more work goes on-line, computers get closer to the heart of business operations. And the cost of downtime goes up."

But even if fault tolerance doesn't take off, Kastner said, he believes that Stratus, "based on our price/performance, can stand on our own two feet."

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positions it against the Prime Computer Inc. 750, DEC VAX 11/750 and IBM 4341 Model 1. The company's under-\$200,000 price tag pits it against 32-bit computers. "We're not directly going after Tandem with price/performance," he said.

But Kastner said he expects other companies to come after Stratus. "By 1986, all [broad-based] computer companies will have to offer fault-tolerant systems," he said. "And at least a half dozen or more will be shipping by the end of 1983."

In a recent meeting of security analysts in Boston, DG President Edson de Castro said com-

patibility," Kastner said, but would not rule out the possibility of a second generation, based on a new technology, that could talk to the older machines via Stratalink. A networking capability built into the operating system, Stratalink uses a proprietary protocol to move information between Stratus machines.

Stratus CPU boards use four 68000 chips, paired off to run operating system code and user application code. Because Stratus writes its software in high-level language, it uses only several thousand lines of code. Therefore, the operating system doesn't have to change one line

operating system makes errors, and downtime, unlikely. Nevertheless, Stratus provides two types of service: standard "if it breaks, we'll fix it" service and remote diagnostics of both hardware and software.

"Rather than Stratus sending out an engineer, who opens the system door and throws in a board, we can send the part — if it fails — over in a cab," Kastner said. As a result, Stratus doesn't need as many engineers as other companies, and can charge annual maintenance fees of 7 percent/year, instead of the industry standard of 10-12 percent.

Turnkey System Aimed at Project Managers

VANCOUVER, B.C. — Aimed at system development managers, project leaders and software consultants, the Turnkey System Planning Model from Augur Data Planning Inc. offers techniques for planning, marketing, estimating and scheduling turnkey computer projects.

Described as a "small project methodology," the package con-

sists of two manuals that describe methods and procedures, plus illustrated diagrams depicting the phases and activities of a standard turnkey project. It specifically addresses projects in the 2-20 work-month range that involve both package and custom software.

According to its developer, Jim Pettinger, use of the model provides "two important benefits for the system implemen-

said, "but the planning techniques must fit the scale of the problem. They must be easy to understand, easy to apply, easy to change and usable for a wide variety of projects in a uniform manner. When all projects are conducted according to this one standard formula, the systems manager has the feeling that he is in control, especially as the company grows and begins add-

ing more people and more projects."

Pettinger, a manager of over 40 small business computer projects, is the author of three manuals on systems and project management.

Augur Data Planning Inc., 1401 West Broadway, Vancouver, BC, V6H 1H6; or P.O. Box 1978, Bellingham, Wash. 98227, (604) 731-1413.